

HPC & AI Practical Viva Questions and Answers

This document contains important viva questions and answers based on your practical programs including OpenMP, Parallel Sorting, BFS/DFS, RNN Stock Prediction, and Fashion MNIST classification.

1. OpenMP Reduction (Min, Max, Sum, Average)

Question	Answer
What is OpenMP?	OpenMP is an API used for parallel programming in C/C++ using multiple threads.
What is reduction in OpenMP?	Reduction combines results from multiple threads into a single variable.
Why use reduction clause?	To avoid race conditions while performing operations like sum, min, and max.
What is race condition?	When multiple threads access shared data simultaneously causing incorrect results.
Why use -fopenmp?	It enables OpenMP support during compilation.
What is parallel for?	It distributes loop iterations among multiple threads.
Why normalize execution time in microseconds?	Microseconds provide better readability for small execution times.

2. Parallel Bubble Sort and Merge Sort

Question	Answer
What is Bubble Sort?	Bubble Sort repeatedly compares adjacent elements and swaps them if needed.
What is Merge Sort?	Merge Sort is a divide-and-conquer sorting algorithm.
Which sorting algorithm is faster?	Merge Sort is faster because it has $O(n \log n)$ complexity.
Why is Bubble Sort slow?	Bubble Sort has $O(n^2)$ time complexity.
Why use even-odd phase in parallel bubble sort?	To avoid conflicts while swapping adjacent elements in parallel.
What is recursion?	A function calling itself repeatedly.
Why parallel version may become slower?	Thread creation and synchronization overhead dominate for small inputs.
What is <code>omp_get_wtime()</code> ?	It measures execution time in OpenMP.

3. Parallel BFS and DFS

Question	Answer
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What is BFS?	Breadth First Search visits graph nodes level by level.
What is DFS?	Depth First Search explores graph depth-wise.
Which data structure is used in BFS?	Queue.
Which data structure is used in DFS?	Stack or recursion.
Why use critical section?	To safely access shared data structures.
What is #pragma omp task?	It creates independent parallel tasks.
Why use visited array?	To avoid revisiting nodes.
What is graph traversal?	Process of visiting all vertices of a graph.

4. Google Stock Price Prediction using RNN

Question	Answer
What is RNN?	Recurrent Neural Network used for sequential/time-series data.
Why use RNN for stock prediction?	Because stock prices depend on previous values.
Why normalize data?	To improve neural network training performance.
What is epoch?	One complete training cycle over the dataset.
What is batch size?	Number of samples processed together.
Why use 60 previous days?	To learn patterns from historical data.
What is MSE?	Mean Squared Error used as loss function.
What is Adam optimizer?	An optimization algorithm used to update neural network weights.

5. Fashion MNIST using Deep Neural Network

Question	Answer
What is Fashion MNIST?	Dataset of clothing images used for image classification.
Why use Flatten layer?	To convert 2D image into 1D vector.
What is ReLU?	Activation function defined as $\max(0, x)$.
Why use Softmax?	To produce probabilities for multiple classes.
What is Dense layer?	Fully connected neural network layer.
What is image classification?	Identifying object category from an image.
What is validation split?	Part of training data reserved for validation.
What is accuracy?	Percentage of correct predictions.

Tip: During viva, explain concepts confidently with practical examples and mention OpenMP directives, neural networks, recursion, and complexity wherever relevant.